```dart
// Automatic FlutterFlow imports
import '/backend/schema/structs/index.dart';
import '/flutter_flow/flutter_flow_theme.dart';
import '/flutter_flow/flutter_flow_util.dart';
import '/custom_code/widgets/index.dart'; // Imports other custom widgets
import 'package:flutter/material.dart';
// Begin custom widget code
// DO NOT REMOVE OR MODIFY THE CODE ABOVE!

import 'dart:convert';
import 'package:flutter/services.dart';
import 'package:http/http.dart' as http;
import 'package:intl/intl.dart';


class ClosureandVerificationAmbience extends StatefulWidget {
  const ClosureandVerificationAmbience({
    super.key,
    this.width,
    this.height,
    this.baseurl,
    this.branchId,
    this.empCode,
  });
  final double? width;
  final double? height;
  final String? baseurl;
  final String? branchId;
  final String? empCode;
  @override
  State<ClosureandVerificationAmbience> createState() =>
      _ClosureandVerificationAmbienceState();
}

class _ClosureandVerificationAmbienceState
    extends State<ClosureandVerificationAmbience> {
  final DateFormat _dateFormat = DateFormat('dd-MM-yyyy');
  List<String> _auditIds = [];
  String? _selectedAuditId;
  List<_AuditRow> _rows = [];

  bool _isLoadingList = false;
  bool _isLoadingDetails = false;
  bool _isSubmitting = false;
  String? _errorMessage;

  // Per-row inputs (same index as _rows)
  // status: 1 = Satisfactory closed, 2 = Wrong Rectification
  List<TextEditingController> _remarkControllers = [];
  List<int> _statuses = [];

  String get _baseUrl => (widget.baseurl ?? '').replaceAll(RegExp(r'/$'), '');

  void _disposeRowControllers() {
    for (final c in _remarkControllers) {
      c.dispose();
    }
    _remarkControllers = [];
    _statuses = [];
  }

  @override
  void dispose() {
```

```
  _disposeRowControllers();
  super.dispose();
}

@override
void initState() {
  super.initState();
  _loadAuditIds();
}

// On load: fetch Audit IDs for the branch
Future<void> _loadAuditIds() async {
  final branchId = widget.branchId?.trim();
  if (_baseUrl.isEmpty || branchId == null || branchId.isEmpty) {
    setState(() => _errorMessage = 'Branch and API URL are required.');
    return;
  }
  setState(() {
    _isLoadingList = true;
    _errorMessage = null;
  });
  try {
    final response = await http.get(Uri.parse(
      '$_baseUrl/AuditInspectionApi/api/GetInspectionData/'
      'Audit_id_ambience/$branchId/1',
    ));
    if (response.statusCode < 200 || response.statusCode >= 300) {
      throw Exception('Unable to load audit ids.');
    }
    final result = _resultList(response.body);
    if (!mounted) return;

    // API returns duplicate AUDIT_IDs, so de-duplicate (keeps order)
    final ids = <String>[];
    for (final row in result) {
      if (row is Map<String, dynamic>) {
        final id = '${row['AUDIT_ID'] ?? ''}'.trim();
        if (id.isNotEmpty && !ids.contains(id)) ids.add(id);
      }
    }
    setState(() {
      _auditIds = ids;
      _isLoadingList = false;
    });
  } catch (_) {
    if (!mounted) return;
    setState(() {
      _isLoadingList = false;
      _errorMessage = 'Unable to load audit id list.';
    });
  }
}

// On select: fetch the datatable for the chosen Audit ID
Future<void> _loadDetails(String auditId) async {
  if (_baseUrl.isEmpty) return;
  setState(() {
    _disposeRowControllers();
    _selectedAuditId = auditId;
    _rows = [];
    _isLoadingDetails = true;
    _errorMessage = null;
  });
  try {
    final response = await http.get(Uri.parse(
      '$_baseUrl/AuditInspectionApi/api/GetInspectionData/'
      'Audit_id_Datatable/$auditId/1',
    ));
    if (response.statusCode < 200 || response.statusCode >= 300) {
```

```
      throw Exception('Unable to load audit data.');
    }
    final rows = _resultList(response.body)
        .whereType<Map<String, dynamic>>()
        .map(_AuditRow.fromJson)
        .toList();
    if (!mounted) return;
    setState(() {
      _rows = rows;
      _remarkControllers =
          List.generate(rows.length, (_) => TextEditingController());
      _statuses = List.filled(rows.length, 1);
      _isLoadingDetails = false;
    });
  } catch (_) {
    if (!mounted) return;
    setState(() {
      _isLoadingDetails = false;
      _errorMessage = 'Unable to load details for the selected audit id.';
    });
  }
}

List<dynamic> _resultList(String body) {
  final decoded = jsonDecode(body);
  return decoded is Map<String, dynamic> && decoded['result'] is List
      ? decoded['result'] as List<dynamic>
      : <dynamic>[];
}

Future<void> _showAlert(String message) async {
  if (!mounted) return;
  await showDialog<void>(
    context: context,
    builder: (context) => AlertDialog(
      title: const Text('Alert'),
      content: Text(message),
      actions: [
        TextButton(
          onPressed: () => Navigator.of(context).pop(),
          child: const Text('OK'),
        ),
      ],
    ),
  );
}

Future<void> _submitClosure(String flag) async {
  final auditId = _selectedAuditId;
  final branchId = widget.branchId?.trim();
  final empCode = widget.empCode?.trim();
  final currentDate = _dateFormat.format(DateTime.now());
  if (auditId == null || branchId == null || branchId.isEmpty) {
    await _showAlert('Please select an audit id first.');
    return;
  }
  if (_rows.isEmpty) {
    await _showAlert('No audit rows to verify.');
    return;
  }
  // every row needs its own remarks
  for (var i = 0; i < _rows.length; i++) {
    if (_remarkControllers[i].text.trim().isEmpty) {
      await _showAlert('Please enter remarks for row ${i + 1} '
          '(${_rows[i].category}).');
      return;
    }
  }
  if (empCode == null || empCode.isEmpty) {
```

```
      setState(() => _errorMessage = 'Employee code is required.');
      return;
    }
    setState(() {
      _isSubmitting = true;
      _errorMessage = null;
    });
    try {
      // One request per row. Category is sent so the backend can tell
      // the rows apart. Confirm this pageValue format with the backend team.
      String responseMessage = 'Successfully updated status.';
      for (var i = 0; i < _rows.length; i++) {
        final remarks = _remarkControllers[i].text.trim();
        final response = await http.post(
          Uri.parse('$_baseUrl/AuditInspectionApi/api/PostInseptionCtl'),
          headers: {'Content-Type': 'application/json'},
          body: jsonEncode({
            'flag': 'closure_and_verify_ambience',
            'pageValue':

'approve*$branchId~$auditId~$empCode~${_statuses[i]}~$remarks~$currentDate~${_rows[i].cate
gory}',
            'paraValue': '1',
          }),
        );
        if (response.statusCode < 200 || response.statusCode >= 300) {
          throw Exception('Unable to submit row ${i + 1}.');
        }
        final decoded = jsonDecode(response.body);
        if (decoded is List &&
            decoded.isNotEmpty &&
            decoded.first is Map<String, dynamic>) {
          final res = '${(decoded.first as Map<String, dynamic>)['RES'] ?? ''}';
          if (res.isNotEmpty) responseMessage = res;
        }
      }
      if (!mounted) return;
      setState(() {
        _isSubmitting = false;
        _errorMessage = null;
        _disposeRowControllers();
        _selectedAuditId = null;
        _rows = [];
      });
      await showDialog<void>(
        context: context,
        builder: (context) => AlertDialog(
          title: Text(responseMessage),
          actions: [
            TextButton(
              onPressed: () => Navigator.of(context).pop(),
              child: const Text('OK'),
            ),
          ],
        ),
      );
      _loadAuditIds();
    } catch (_) {
      if (!mounted) return;
      setState(() {
        _isSubmitting = false;
        _errorMessage = 'Unable to submit the closure action.';
      });
    }
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
```

```
    backgroundColor: const Color(0xFFF7F8FA),
    appBar: AppBar(
      leading: IconButton(
        tooltip: 'Back',
        icon: const Icon(Icons.arrow_back, color: Color(0xFF1A202C)),
        onPressed: () => Navigator.of(context).maybePop(),
      ),
      centerTitle: true,
      title: const Text(
        'Security & Ambiance Closure',
        style: TextStyle(
          fontWeight: FontWeight.w600,
          fontSize: 20,
          color: Color(0xFF1A202C),
        ),
      ),
      backgroundColor: Colors.white,
      surfaceTintColor: Colors.transparent,
      shadowColor: Colors.black12,
    ),
    body: Row(
      crossAxisAlignment: CrossAxisAlignment.start,
      children: [
        Expanded(
          child: SingleChildScrollView(
            padding: const EdgeInsets.symmetric(
              horizontal: 48.0,
              vertical: 32.0,
            ),
            child: Column(
              crossAxisAlignment: CrossAxisAlignment.stretch,
              children: [
                if (_errorMessage != null)
                  Padding(
                    padding: const EdgeInsets.only(bottom: 16.0),
                    child: Text(
                      _errorMessage!,
                      style: TextStyle(
                        color: Colors.red.shade700,
                        fontWeight: FontWeight.w600,
                      ),
                    ),
                  ),
                _buildAuditSelectionSection(),
                if (_isLoadingDetails)
                  const Padding(
                    padding: EdgeInsets.symmetric(vertical: 40),
                    child: Center(
                      child: CircularProgressIndicator(
                        color: Color(0xFF005FB8),
                      ),
                    ),
                  ),
                if (_selectedAuditId != null && !_isLoadingDetails) ...[
                  const SizedBox(height: 32),
                  _buildAuditTableSection(),
                  const SizedBox(height: 24),
                  _buildActionButtons(),
                  const SizedBox(height: 32),
                ],
              ],
            ),
          ),
        ),
      ],
    ),
  );
}
```

```
Widget _buildAuditSelectionSection() {
  return _SectionContainer(
    title: 'Audit Selection',
    icon: Icons.assignment_ind_outlined,
    child: Row(
      mainAxisAlignment: MainAxisAlignment.start,
      children: [
        _FormDropdown<String>(
          label: 'Select Audit ID',
          width: 260,
          value: _selectedAuditId,
          items: _auditIds,
          onChanged: _isLoadingList
              ? null
              : (value) {
                  if (value != null) _loadDetails(value);
                },
          displayText: (item) => item,
          hintText: _isLoadingList ? 'Loading...' : 'Select...',
        ),
      ],
    ),
  );
}

Widget _buildAuditTableSection() {
  const headStyle = TextStyle(
    fontSize: 12,
    fontWeight: FontWeight.w700,
    color: Color(0xFF2D3748),
  );
  const cellStyle = TextStyle(fontSize: 12, color: Color(0xFF1A202C));

  Widget cell(String text, double w) => ConstrainedBox(
        constraints: BoxConstraints(maxWidth: w),
        child: Text(text.isEmpty ? '-' : text, style: cellStyle),
      );

  return _SectionContainer(
    title: 'Audit Details',
    icon: Icons.table_chart_outlined,
    child: _rows.isEmpty
        ? const Padding(
            padding: EdgeInsets.symmetric(vertical: 16),
            child: Center(
              child: Text(
                'No data found for this audit id.',
                style: TextStyle(color: Color(0xFF4A5568)),
              ),
            ),
          )
        : LayoutBuilder(
            builder: (context, constraints) => SingleChildScrollView(
              scrollDirection: Axis.horizontal,
              child: ConstrainedBox(
                constraints: BoxConstraints(minWidth: constraints.maxWidth),
                child: DataTable(
                  headingRowColor:
                      WidgetStateProperty.all(const Color(0xFFF9FAFB)),
                  dataRowMinHeight: 40,
                  dataRowMaxHeight: 72,
                  columnSpacing: 32,
                  border: const TableBorder(
                    horizontalInside:
                        BorderSide(color: Color(0xFFEDF2F7)),
                  ),
                  columns: const [
                    DataColumn(label: Text('Sr.', style: headStyle)),
                    DataColumn(label: Text('Category', style: headStyle)),
```

```
                    DataColumn(label: Text('Question', style: headStyle)),
                    DataColumn(
                        label: Text('Audit Remark', style: headStyle)),
                    DataColumn(label: Text('BH Answer', style: headStyle)),
                    DataColumn(
                        label: Text('Auditor Remarks', style: headStyle)),
                    DataColumn(
                        label:
                            Text('Verification Status', style: headStyle)),
                  ],
                  rows: [
                    for (var i = 0; i < _rows.length; i++)
                      DataRow(cells: [
                        DataCell(Text('${i + 1}', style: cellStyle)),
                        DataCell(cell(_rows[i].category, 180)),
                        DataCell(cell(_rows[i].question, 380)),
                        DataCell(cell(_rows[i].auditorRemark, 220)),
                        DataCell(cell(_rows[i].bhAnswer, 160)),
                        DataCell(_remarkField(i)),
                        DataCell(_statusDropdown(i)),
                      ]),
                  ],
                ),
              ),
            ),
          ),
        ),
    );
  }

  Widget _remarkField(int i) {
    return SizedBox(
      width: 260,
      height: 36,
      child: TextField(
        controller: _remarkControllers[i],
        style: const TextStyle(fontSize: 12, color: Colors.black),
        decoration: InputDecoration(
          hintText: 'Enter remarks...',
          hintStyle: const TextStyle(fontSize: 12, color: Colors.grey),
          filled: true,
          fillColor: Colors.white,
          contentPadding: const EdgeInsets.symmetric(horizontal: 10),
          border: OutlineInputBorder(
            borderRadius: BorderRadius.circular(6),
            borderSide: const BorderSide(color: Color(0xFFCBD5E0)),
          ),
          enabledBorder: OutlineInputBorder(
            borderRadius: BorderRadius.circular(6),
            borderSide: const BorderSide(color: Color(0xFFCBD5E0)),
          ),
          focusedBorder: OutlineInputBorder(
            borderRadius: BorderRadius.circular(6),
            borderSide: const BorderSide(color: Color(0xFF005FB8), width: 1.5),
          ),
        ),
      ),
    );
  }

  Widget _statusDropdown(int i) {
    return Container(
      width: 200,
      height: 36,
      padding: const EdgeInsets.symmetric(horizontal: 10),
      decoration: BoxDecoration(
        color: Colors.white,
        borderRadius: BorderRadius.circular(6),
        border: Border.all(color: const Color(0xFFCBD5E0)),
      ),
```

```
      child: DropdownButtonHideUnderline(
        child: DropdownButton<int>(
          value: _statuses[i],
          isExpanded: true,
          icon: const Icon(Icons.arrow_drop_down, color: Color(0xFF4A5568)),
          style: const TextStyle(fontSize: 12, color: Color(0xFF1A202C)),
          items: const [
            DropdownMenuItem(value: 1, child: Text('Satisfactory closed')),
            DropdownMenuItem(value: 2, child: Text('Wrong Rectification')),
          ],
          onChanged: (value) {
            if (value != null) setState(() => _statuses[i] = value);
          },
        ),
      ),
    );
  }

  Widget _buildActionButtons() {
    return Row(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        ElevatedButton.icon(
          onPressed: _isSubmitting ? null : () => _submitClosure('0'),
          icon: const Icon(Icons.check),
          label: const Text('Confirm Validation'),
          style: ElevatedButton.styleFrom(
            backgroundColor: const Color(0xFF16803C),
            foregroundColor: Colors.white,
            padding: const EdgeInsets.symmetric(horizontal: 24, vertical: 12),
            shape: RoundedRectangleBorder(
              borderRadius: BorderRadius.circular(6),
            ),
          ),
        ),
        const SizedBox(width: 16),
        OutlinedButton.icon(
          onPressed: _isSubmitting ? null : () => context.go('/menuPage'),
          icon: const Icon(Icons.exit_to_app),
          label: const Text('Exit'),
          style: OutlinedButton.styleFrom(
            foregroundColor: const Color(0xFF2D3748),
            padding: const EdgeInsets.symmetric(horizontal: 24, vertical: 12),
            side: const BorderSide(color: Color(0xFFCBD5E0)),
            shape: RoundedRectangleBorder(
              borderRadius: BorderRadius.circular(6),
            ),
          ),
        ),
      ],
    );
  }
}

// --- Reusable UI Components
class _SectionContainer extends StatelessWidget {
  final String title;
  final IconData icon;
  final Widget child;
  final EdgeInsetsGeometry padding;
  const _SectionContainer({
    required this.title,
    required this.icon,
    required this.child,
    this.padding = const EdgeInsets.all(24.0),
  });
  @override
  Widget build(BuildContext context) {
    return Container(
```

```
      decoration: BoxDecoration(
        color: Colors.white,
        borderRadius: BorderRadius.circular(12),
        boxShadow: [
          BoxShadow(
            color: Colors.black.withOpacity(0.05),
            blurRadius: 20,
            offset: const Offset(0, 5),
          ),
        ],
        border: Border.all(color: const Color(0xFFEDF2F7)),
      ),
      child: Column(
        children: [
          const SizedBox(height: 8),
          Container(
            padding: const EdgeInsets.symmetric(
              horizontal: 24.0,
              vertical: 16.0,
            ),
            decoration: const BoxDecoration(
              color: Color(0xFFF9FAFB),
              borderRadius: BorderRadius.only(
                topLeft: Radius.circular(12),
                topRight: Radius.circular(12),
              ),
              border: Border(
                bottom: BorderSide(color: Color(0xFFEDF2F7)),
              ),
            ),
            child: Row(
              children: [
                Icon(
                  icon,
                  size: 22,
                  color: const Color(0xFF4A5568),
                ),
                const SizedBox(width: 12),
                Text(
                  title,
                  style: const TextStyle(
                    fontSize: 16,
                    fontWeight: FontWeight.w600,
                    color: Color(0xFF2D3748),
                  ),
                ),
              ],
            ),
          ),
          Padding(
            padding: padding,
            child: child,
          ),
        ],
      ),
    );
  }
}

class _FormDropdown<T> extends StatelessWidget {
  final String label;
  final double width;
  final T? value;
  final List<T> items;
  final void Function(T?)? onChanged;
  final String Function(T) displayText;
  final String hintText;
  const _FormDropdown({
    Key? key,
```

```
  required this.label,
  required this.width,
  required this.value,
  required this.items,
  required this.onChanged,
  required this.displayText,
  this.hintText = 'Select...',
}) : super(key: key);
@override
Widget build(BuildContext context) {
  return SizedBox(
    width: width,
    child: Column(
      crossAxisAlignment: CrossAxisAlignment.start,
      children: [
        Text(
          label,
          style: const TextStyle(
            fontSize: 11,
            fontWeight: FontWeight.w600,
            color: Color(0xFF4A5568),
          ),
        ),
        const SizedBox(height: 8),
        Container(
          height: 36,
          padding: const EdgeInsets.symmetric(horizontal: 10),
          decoration: BoxDecoration(
            color: Colors.white,
            borderRadius: BorderRadius.circular(6),
            border: Border.all(color: const Color(0xFFCBD5E0)),
            boxShadow: [
              BoxShadow(
                color: Colors.black.withOpacity(0.02),
                blurRadius: 4,
                offset: const Offset(0, 2),
              ),
            ],
          ),
          child: DropdownButtonHideUnderline(
            child: DropdownButton<T>(
              hint: Text(
                hintText,
                style: const TextStyle(fontSize: 12, color: Colors.grey),
              ),
              value: value,
              isExpanded: true,
              icon:
                  const Icon(Icons.arrow_drop_down, color: Color(0xFF4A5568)),
              items: items.map((item) {
                return DropdownMenuItem<T>(
                  value: item,
                  child: Text(
                    displayText(item),
                    style: const TextStyle(fontSize: 12),
                    overflow: TextOverflow.ellipsis,
                  ),
                );
              }).toList(),
              onChanged: onChanged,
              style: const TextStyle(
                fontSize: 12,
                color: Color(0xFF1A202C),
              ),
            ),
          ),
        ),
      ],
    ),
```

```
    );
  }
}

// --- Data Models
class _AuditRow {
  const _AuditRow({
    required this.category,
    required this.question,
    required this.auditorRemark,
    required this.bhAnswer,
  });
  final String category;
  final String question;
  final String auditorRemark;
  final String bhAnswer;
  factory _AuditRow.fromJson(Map<String, dynamic> json) => _AuditRow(
        category: '${json['CATEGORY'] ?? ''}',
        question: '${json['QUESTION'] ?? ''}',
        auditorRemark: '${json['AUDITOR_REMARK'] ?? ''}',
        bhAnswer: '${json['BH_ANSWER'] ?? ''}',
      );
}
```